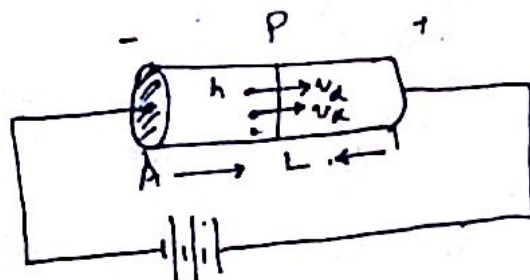


①

Electrical Conductivity

- ① When electrons are almost free.
- ② Motion of electrons in the metal is random.
- ③ Free electrons during random motion collides with each other.
- ④ Kinetic theory of Gas laws are applicable.

Conductivity ::



L — Length of conductor
 n — Electron density.
 v_d — Average drift speed.

$\therefore$ No. of electrons crossing any plane P of cross-sectional area A in 1 sec
 where $n = \text{no. of electrons/vol.}$
 $= n A \times v_d$

$\therefore$ charge carried by electrons/sec = $n A e v_d$ ✓

$$\therefore I = n A e v_d \quad \text{--- (1)}$$

$$\therefore J = \frac{I}{A} = n e v_d \quad \text{--- (2)}$$

Also (2) can be written as

$$J = \rho \cdot v_d \quad \text{--- (3) where } \rho = n e \text{ is charge/vol.}$$

Also we know that

$$v_d \propto E \quad \text{or} \quad v_d = \mu_e E \quad \text{where } \mu_e \text{ is called electron mobility.} \quad \text{--- (4)}$$

put V_d in (2)

$$\therefore J = ne\mu_e E$$

or $J = \sigma E$ — (5) where $\sigma = ne\mu_e$ where σ is called the conductivity of the conductor and has units ohm⁻¹m⁻¹.
 sp. resistance resistivity $\rho = \frac{1}{\text{Conductivity}} = \frac{1}{ne\mu_e}$

Conductivity of Semi Conductors :- The conductivity of semiconductors is different from metals in respect that in metals — only the electrons are charge carriers while in intrinsic semi conductors both electrons & holes are charge carriers (Bipolar) while metals are unipolar.

$\sigma = ne\mu_e$
 $\therefore$ conductivity in metals ~~XXXXXXXXXXXX~~ — (1)
 where σ is conductivity & $\mu_e = \text{Mobility of electrons}$

In Semi Conductors

let $n = \text{electron density / cm}^3$ or $\frac{\text{no of electrons in conduction band}}{\text{Vol.}}$

$p = \text{positive hole density in valence band}$

$\mu_e = \text{Electron mobility}$

$\mu_h = \text{Hole mobility}$

$\therefore \sigma_n = ne\mu_e$ in conduction band (electrons)

$\sigma_p = pe\mu_h$ in valence band (Holes)

$$\therefore \sigma = \sigma_n + \sigma_p$$

$$\therefore \sigma = ne\mu_e + pe\mu_h = e[n\mu_e + p\mu_h] \quad \text{--- (2)}$$

Intrinsic Semi Conductors :-

pure Ge & Si

$$\therefore n = p = n_i$$

$$\therefore \sigma_i = e [n_i \mu_e + n_i \mu_h]$$

$$\sigma_i = e n_i [\mu_e + \mu_h]$$

N type Semi Conductor Conductivity

when a small quantity of Pentavalent Impurity such as Phosphorus is added — resulting crystal is N-type extrinsic semi-conductor.

In $n_e \gg n_h$ or p .

$\therefore$ The electron concentration is denoted by N_D (conc. of donor atom)

$$\therefore \sigma_n \approx e N_D \mu_e \quad (\text{as } n \gg p)$$

P-type $\therefore$ when a trivalent Impurity is added such as Indium or Boron.

$$\therefore p \gg n.$$

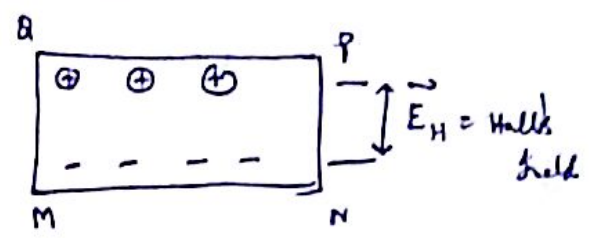
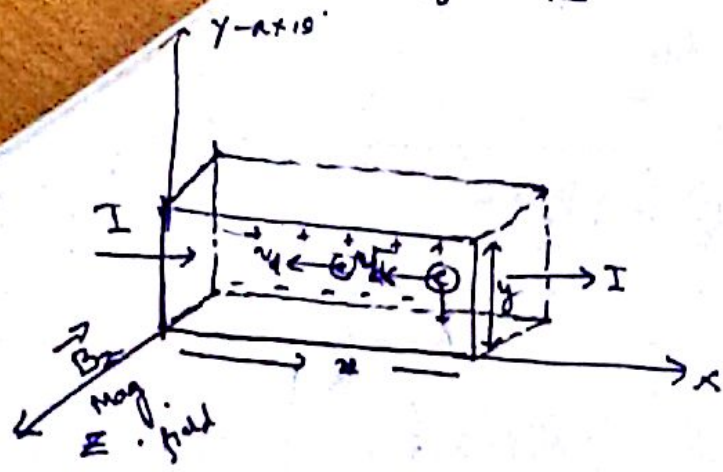
Let conc. is represented by N_A i.e., conc. of Acceptor atoms.

$\therefore$ conductivity of P-type semiconductor $\hookrightarrow$

$$\sigma_p \approx e N_A \mu_h \quad \therefore \text{proof} \quad (p \gg n)$$

Halls effect :-

E.H. Hall in 1879 observed this effect.



* $\vec{L} \propto \vec{B}$

Force on the conduction electron due to electric field $= -eE_H$

due to Magnetic field $= -ev_d B_z$

In Equilibrium

$$eE_H = ev_d B_z$$

$$\boxed{26 \text{ } v_d = v_{sc}}$$

$\therefore E_H = v_d B_z$ — (1)

(Note here $v_x = v_d$)

Current density in x-direction $J_x = -nev_d$

$$\text{or } v_d = -J_x / ne \quad \text{--- (2)}$$

put v_d in (2) into (1)

$$\therefore E_H = - \left(\frac{J_x}{ne} \right) B_z = - \left(\frac{1}{ne} \right) J_x B_z \quad \text{--- (3)}$$

eq (3) shows that Hall's field is proportional to both current density J_x & B_z .

For (3)

$$\boxed{\frac{E_H}{J_x B_z} = - \frac{1}{ne} = R_H} \quad \text{--- (4)}$$

R_H is called Hall's coefficient

If y and z be thickness and width in y - & z - directions

$$\therefore E_H = \left(\frac{V_H}{y} \right) \quad \text{where } V_H = \text{Hall's voltage}$$

For (3)

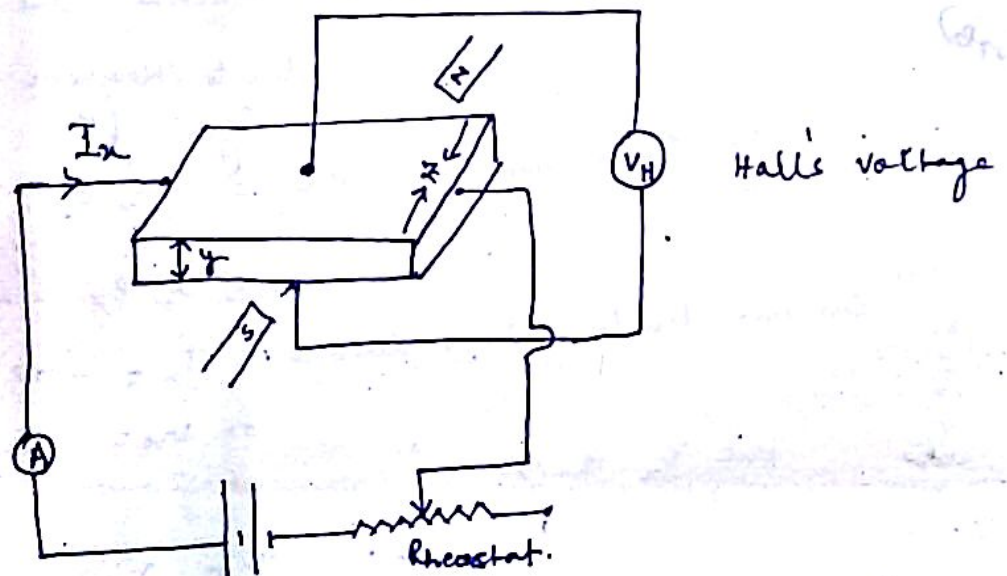
$$\therefore \frac{(V_H/y)}{J_x B_z} = R_H \quad \text{or } \boxed{V_H = J_x B_z y R_H}$$

$$\text{Again } J_x = \frac{I_x}{y \cdot z} \quad \therefore V_H = \left(\frac{I_x}{y \cdot z} \right) B_z y R_H$$

$$\text{or } V_H = \frac{I_x B_z R_H}{z}$$

Hence $R_H = \text{Hall's coeff} = \frac{V_H \cdot z}{I_x B_z}$ VVSP

Experimental Measurement of R_H (Hall's coeff)



Current density $J = \frac{I_x}{A} = \frac{I_x}{yz}$

Importance

- ① It shows that in metals, electrons are used in electrical conduction.
- ② n can be determined
- ③ Mobility μ_e & μ_h can be determined